

Document Q&A; System Architecture Design

Problem Statement

The goal is to build a scalable Document Q&A; system over 10,000 internal documents including PDFs, DOCX files, and Confluence pages. The system should support semantic search, accurate retrieval, citation-based answers, and enterprise scalability.

High-Level Architecture

Documents → Parsing & Cleaning → Chunking → Embedding Generation → Vector Database → Hybrid Retrieval → Re-ranking → LLM Answer Generation → Cited Response

Document Ingestion & Processing

The system supports PDFs, DOCX files, and Confluence pages. PDFs are parsed using PyPDFLoader, DOCX files using Docx2txtLoader, and Confluence pages through API-based ingestion.

Data Cleaning

The preprocessing pipeline removes headers, footers, duplicate spaces, and formatting noise while preserving metadata such as document source, page number, and section title.

Chunking Strategy

Recursive Semantic Chunking with overlap is used. Chunk size is 500–800 tokens with 100–150 token overlap. This preserves contextual continuity and improves retrieval accuracy.

Embedding Model Choice

The system uses OpenAI text-embedding-3-large for semantic embeddings. Embeddings convert text into vectors that help semantic similarity search.

Vector Database Choice

ChromaDB is used during development and Pinecone for production scalability. Vector databases support Approximate Nearest Neighbor (ANN) search and metadata filtering.

Hybrid Retrieval

The retrieval layer combines semantic vector search with BM25 keyword search. Semantic search captures intent while keyword search handles exact terms such as IDs and error codes.

Re-ranking

Retrieved chunks are passed through a Cross-Encoder Re-ranker to improve precision and remove noisy retrieval results.

Handling Partial Answers

Some answers span multiple chunks or documents. The system retrieves top-k chunks, merges contexts, removes duplicate information, and sends aggregated context to the LLM.

LLM Answer Generation

GPT-4.1 generates final answers using only retrieved context. The model is instructed to avoid hallucinations and provide citations.

Citation System

Each chunk stores metadata such as source file, page number, and document ID. The final response includes citations for trust and explainability.

Biggest Failure Mode

The biggest failure mode is retrieval failure, where relevant chunks are not retrieved. This may lead to incomplete or hallucinated answers.

Mitigation Strategies

Retrieval failure is mitigated using hybrid retrieval, query expansion, re-ranking, better chunking, confidence thresholds, and parent-child retrieval.

Scalability Considerations

For 10,000+ documents, the system uses ANN indexing, batch embedding generation, distributed vector databases, caching, and asynchronous ingestion pipelines.

Production Improvements

Production systems should include authentication, monitoring, analytics, logging, caching, document versioning, and incremental indexing.

Final Summary

This architecture provides semantic understanding, accurate retrieval, citation-based responses, reduced hallucinations, and enterprise-grade scalability.